AUDIO SYSTEM

OVERVIEW

The Audio System is primarily controlled by the AudioManager.cs script. It also heavily relies on AudioResourceMap.csv and AudioData.cs. The audio manager controls the playing of all sounds. AudioData is a class that stores a sound clip, a sound clip ID, and what events the sound should be played/stopped on. The audio resource map is a text file that records which sounds should be played on which events.

The audio manager sits on an audio manager game object and always ensures that there are enough AudioSource components attached to it to play all the sounds it needs to play, creating more as needed.

Right when the game is started, the Audio Manager goes into the AudioResourceMap.csv file. For each sound file listed in the resource map, it loads the sound effect and its ID into an AudioData object so that it can later find the sound for an associated sound ID. Then, it adds a listener for the event on which the sound should be played that will call PlaySound(sound ID). That way, the sound will be played whenever the event is called. It also adds a listener for the event that should stop the sound if the sound is continuous.

HOW TO USE THE AUDIO RESOURCE MAP

Each line in the audio resource map is a sound effect. Here is what a line might look like and what each part means.

Example Line:

00001,Place Tile,PlaceTile,0,TilePlacedWithoutDrag,0,none,none

What Each Part Means:

SoundID, Description, Sound File Name, Sound Effect Is Enabled (1 or 0), Event To Play Sound On (See a list in EventType.cs), Sound Effect Loops (1 or 0), Event To End Sound On (Only include if sound effect loops), Secondary Event To Stop Sound

Note that the sound file name shouldn't include the file extension. For example, if the sound file's name is "PlaceTile.mp3", you should write that in the resource map file as "PlaceTile"

HOW TO ADD A SOUND EFFECT

1. Put the new audio file in the Resources/Audio folder so it's packed at runtime (mp3 format works best).
2. Add a new line in the AudioResourceMap.csv (Also located in the Resources/Audio folder) based on the description in the previous section.

Note: You may have to add a new event for the sound effect to played on in the EventType.cs file if one doesn't already exist for when you want the sound effect to be played. For information on adding new events, see "Event_System.pdf"

INTERNAL API

These are some of the most important public functions from the AudioManager.cs script.

void PlaySound(int soundID)

This function plays the sounds associated with the given sound ID. It stops playing the sound once the duration of the sound file has elapsed.

void PlaySoundContinuous(int soundID)

Plays the sound associated with the given sound ID on loop, until EndSound() for the sound ID is called.

void EndSound(int soundID)

Stops playing the sound associated with the given sound ID.

AudioClip GetAudioClipByID(int soundID)

Returns the audio file associated with the given sound ID by looking through the array of AudioData objects for the matching sound ID.

void DoubleNumberOfAudioSources()

This is called when there aren't enough available audio sources to play all of the necessary sounds. It doubles the current number of audio sources.

private void LoadAudioResourceMap()

This is a private function, but it felt helpful to give a description anyway. It's an absurdly overcomplicated function that goes into the AudioResourceMap.csv file and stores the data from it into an array of AudioData objects. This function could be majorly improved and simplified. (If anyone ever has the time to do that.)

-Graydon Bruyn, May 2026